

FLIGHT

Performance & Analysis

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UFMFFK-15-2 Flight

Deliverables, Task and POH

- I was tasked with completing the performance and stability analysis of a regional transport type aircraft.
- Using MATLAB I calculated the take off to the landing of the aircraft and created a relevant POH Document.
- I also analysed the stability of the aircraft at a longitudinal and a lateral flight mode with Simulink.
- Centre of Gravity Limits
- Take off Distance Table
- Rate of Climb Table
- Cruise Performance Table
- Landing Table
- Endurance Profile



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Regional Air freight aircraft POH

Specification

Measurements	Value
Fuselage Length	24.2m
Wingspan	19.25m
Wing Area	32.51m2
Cabin length	14.1m
Performance	
OEW	8812kg
MTOW	16020kg
Max Payload	5290kg
Max Fuel	3600kg
Design Fuel	1918kg
Fuel	Assumed equally distributed in the wing area
MLW	15570kg

Engines	Value
Weight	2x450kg
Engine power	1625kW per engine ISA at sea level
Engine Specific fuel consumption	0.485 lb/hp/hr
Static Thrust per engine	18550N ISA
Propeller Diameter	3.5m, variable pitch
Propeller efficiency	Function of Mach Number

Aircraft masses	Value
Wings	1573.6kg
Horizontal Tail	334.5kg
Vertical fin	289.9kg
Fuselage	5714kg

Landing Gear Loading limits	Value
Front	370kg
Main	18000kg

Parameter	Value
Datum	Tip of root chord at centreline
Crew	2
OEW c.g	1.525m aft of datum

Vertical Tail

Parameter	Value
chord profile	NACA-0015
Rudder	35% local Chord
Taper Ratio	0.6
Max Deflection	+-20°

Wings

Parameter	Value
Wing chord profile	64-210
Rigging Angle	2° relative to fuselage centreline
Washout (root to tip)	2°
High Lift devices	Plain Flaps 35° Max deflection 30% of local Chord. Take-off setting 20°
Dihedral	0°

Horizontal Tail

Parameter	Value
chord profile	NACA-0012
Rigging angle	0° relative to fuselage centreline
Elevator	50% of Tail plane area
Dihedral	0°
Gearing Ratio	3.5rad/m
Max Deflection	+-20°

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COG – Centre of Gravity Limits

- The COG is the location on the aircraft where its weight is balanced.
- The COG contributes to the stability and control of the aircraft whilst in use.
- To calculate the COG I found the front and rear gear locations on the x – axis.
- Once I found these values I can use the this equation:

$$xcg=((369.99*(xfg-xrg)/weight)+xrg)-xdatum$$

- This equation calculates the COG on the x – axis at different weights and plotted the results.

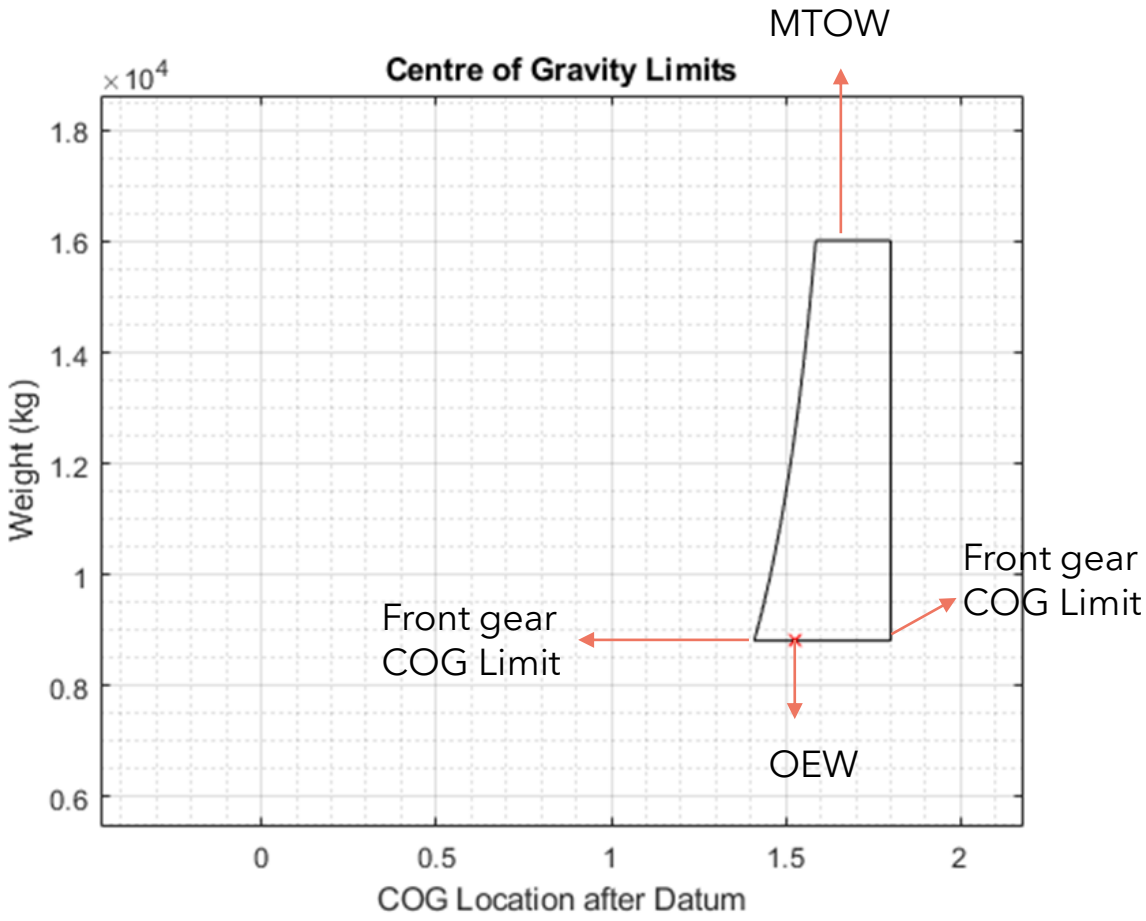


Figure 1

Take-Off Distance Table

- The aircraft we used a NACA 64-210 air foil with an angle of 2 degrees , a flap deflection of 20 degrees. (as part of the take of settings),and a flap chord of 35 degrees.
- Using Java foil I was able to find the maximum lift and drag coefficients which I then converted into 3D coefficients.
- To calculate the total take off distance I need to calculate the Ground run and the air phase part of the take off process.

- To calculate the ground run I used the equation:

$S_G=-W/g\boldsymbol{\rho}S*1/((C_(D_(0,G)\ )-\mu C_(L_G\ )\ )*\ln[1-(C_(D_(0,G)\ )-\mu C_(L_G\ ))/((T/W-\mu)\ C_(L_ROT\ )\ )]$

- To calculate the Air phase I used the equation:

$Airphase=(((MTOW/wingarea)*LiftoffA)/((speed\ of\ sound)*g*Cldelta))+(takeoff\ clearance\ height)/LiftoffA$

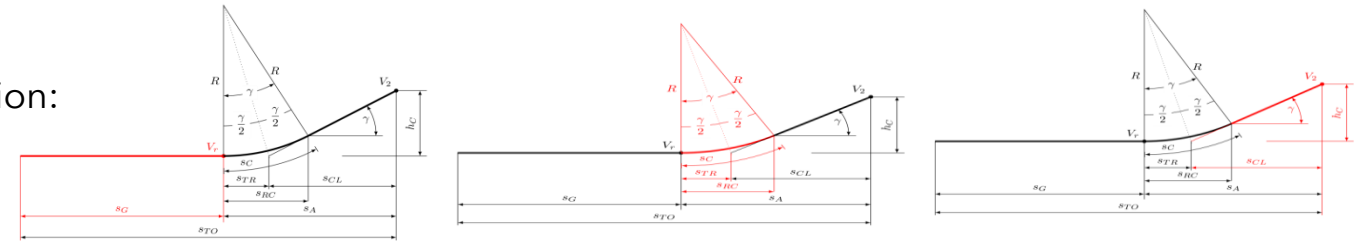
- To calculate the Total take off distance I used the equation:

Total Take-off distance = S_

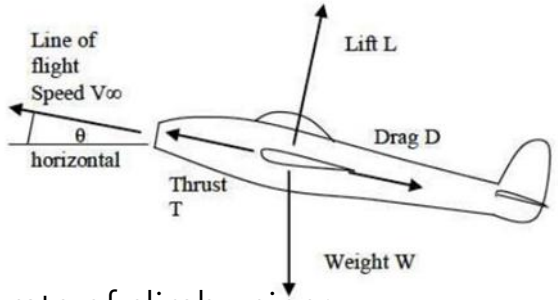
- Due to many airports being above ground I decided to the Take off distance at 5 different altitudes ranging from sea level to 8000ft above sea level.
- From the results I was able to produce table 1 that demonstrate how far it would take for this aircraft to Take off
- To improve I could have found he take off at different cargo weight instead of the MTOW.

Take off MTOW (16020kg)			
Altitude (ft)	Ground Run (ft)	Lift air Phase (ft)	Total Take off Distance (ft)
0 (Sea Level)	2.6147e+03	1.866e+03	9.037e+03
2000	7.6060e+03	1.9792e+03	9.585e+03
4000	8.0743e+03	2.1010e+03	1.018e+04
6000	8.5785e+03	2.2322e+03	1.081e+04
8000	9.1220e+03	2.3736e+03	1.1496e+04

Table 1



Rate of Climb



- Rate of climb refers to the vertical speed at which an aircraft is able to ascend
- As the plane starts to clime the flaps deflection will return to 0 degrees. So I needed to re calculate the lift and drag coefficients in java foil.
- The Drag climb was initially calculated with:

$drag\ climb$=0.5$*air\ density*$($velocity\ climb$.^2) $*climbCd*$wingarea

- Then the Mach climb number :

M$ach\ climb$=$velocity\ climb$/($speed\ of\ sound$)

- Then the Propellor efficiency:

$$\eta = 2.97984M - 6.92675M^2 + 19.1836M^3 - 20.94595M^4$$

- Then the Thrust available:

$thrust\ available$=($propellorEclimb$*(($air\ density$)/($air\ density$ at sea level))$*power$)

- From results I could calculate the rate of climb using:

ROC = ((($thrust\ available\ climb$)−$drag\ climb$)$*velocity\ climb$)/($g*MTOW$)

- I also calculated the rate of climb at different altitudes if the aircraft was to climb from above sea level then the pilot would recognise where they may need to make adjustments.
- Figure 2 shows us the trend of the rate of climb at MTOW (16020kg)

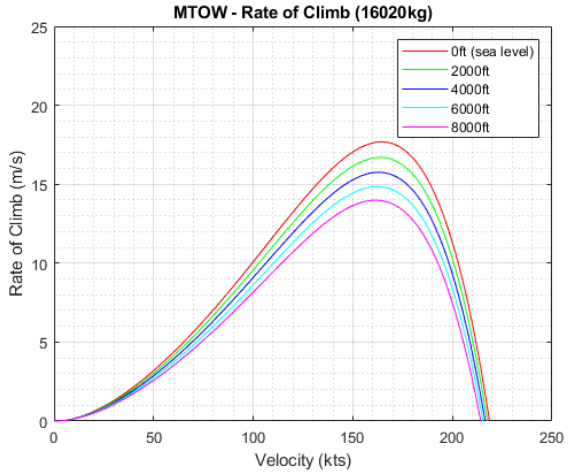


Figure 2

Cruise Performance Table

- Cruise performance refers to the aircraft's ability to maintain a desired airspeed and altitude during the cruise phase of flight. It is influenced by factors such as aircraft weight, atmospheric conditions, engine thrust, and aerodynamic drag.
- As there was not a specific velocity at our cruise I decided to create a range between 0 m/s to 300m/s which increased in 1000 increments.
- The altitude has increased and now at12000ft to 20000ft.
- Due to the increase in altitude the Mach number will change as the air density will decrease.
- To find the optimum Cruise performance I need to compare the Power available compared to the power required.
- To find the power available I used:

$PowerA=(propellerEcruise*(((air\ density)/(air\ density))*power))$

- To find the Power required to cruise I used:

$PowerR=0.5*air\ density*([[velocity\ cruise]]\text{^}3)*wingarea*Cdcruise+(k*([[MOTW]]\text{^}2/((0.5*air\ density*velocity\ cruise*wingarea))))$

- I also wanted to compare the cruise performance at MTOW if the throttle was at different percentages. I chose to do the throttle percentage at 100%, 90% and 80%.
- As shown in Figure 3 we can see where the power available (dotted lines) and the power required intersects.
- The second point at which theses lines intersect it tells us the maximum velocity (kts) where power is substantial.
- If two of the same coloured lines do not intersect then it would mean the aircraft does not have enough power to cruise at the certain altitude thus meaning the plane will start to descend to an altitude it can cruse at.

Cruise at 90% Throttle		
Altitudes (ft)	**Cruise velocity (kts)**	**Fuel consumption (kg/hr/W)**
12000	209.309	1037320
14000	206.927	991063
16000	203.864	959046
18000	200.46	931142
20000	196.717	905713

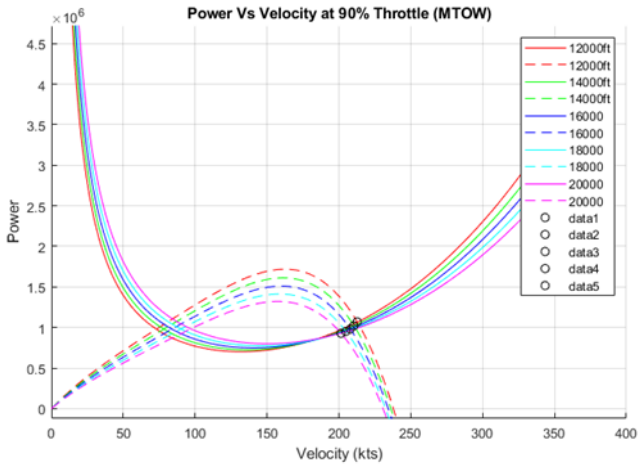


Figure 3

Landing Performance

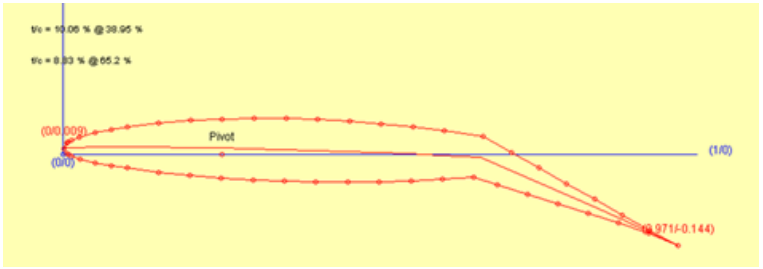
- Landing performance refers to the aircraft's ability to safely land on a runway while meeting certain performance criteria such as landing distance and approach speed. It is influenced by factors such as aircraft weight, approach angle, runway length, atmospheric conditions, and braking capability.
- The Landing is very similar to the take off it is the opposite.
- Using the same NACA 64-210 air foil as take off with an angle of 2 degrees, a flap chord of 35 degrees but a , a flap deflection of 35 degrees
- This meant I needed to use javafoil again to calculate the new lift and drag coefficients.
- I wanted to calculate the maximum total landing to do this I needed to calculate the ground deceleration distance and he landing distance air phase.
- To calculate the ground deceleration distance I used:

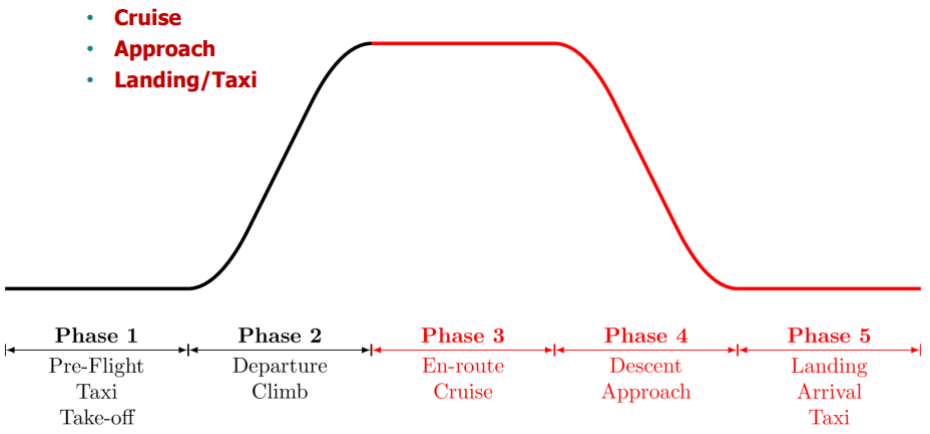
$$sg = Vland\ 2\ 2a = Vland\ 2\ 2g\mu$$

- To calculate the landing distance air phase I used:

$$airphaselanding = clearance_height * (landingt/Deff + ((\llbracket Apvelocity \rrbracket \hat{}2 - (\llbracket landvelocity \rrbracket \hat{}2\ ))/(2 * g))$$

- From these two values you can calculate the total landing distance by finding the sum of them both.





Endurance Profile

- The endurance profile of an aircraft refers to its ability to remain airborne for a specified period of time, determined by the amount of fuel on board and the fuel consumption rate. It is affected by factors such as the aircraft's weight, engine efficiency, altitude, and airspeed.
- I calculated the endurance at the different throttle percentages to demonstrate whether the aircraft will last longer or short time accelerating.
- the endurance is determined by the design fuel/fuel consumption.
- I calculated this at 5 different altitudes and plotted it on figure 4, figure 5 and figure 6. Each figure is at different throttle percentages and show that the aircraft's endures is under 7 hours which is realistic for the aircraft size.

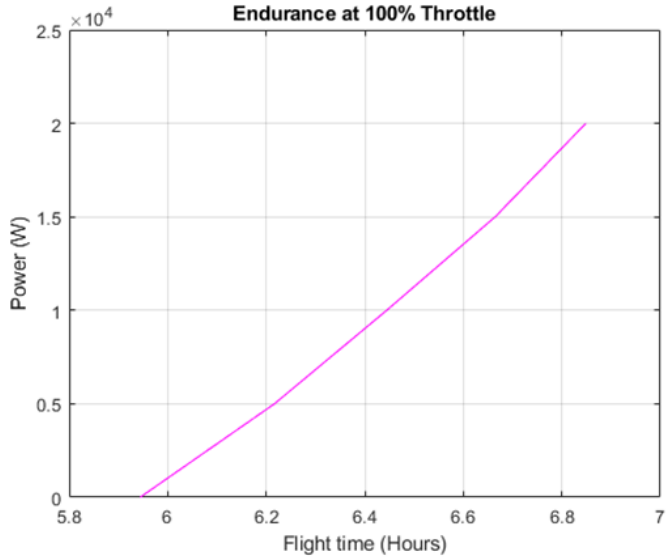


Figure 4

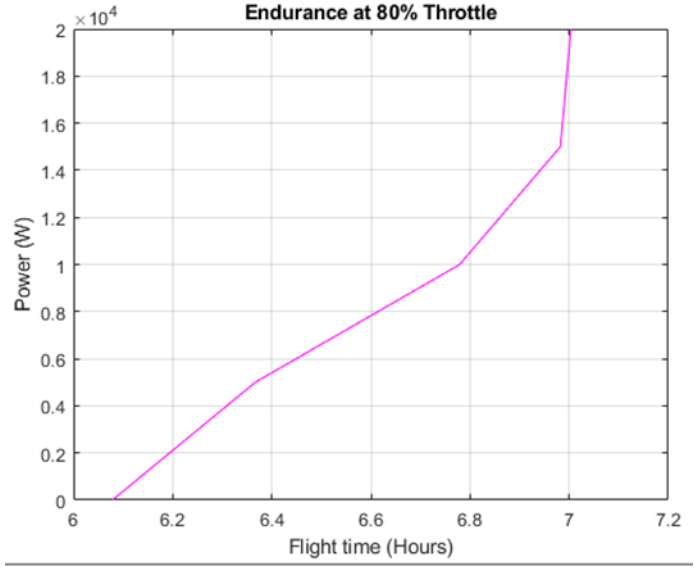


Figure 6

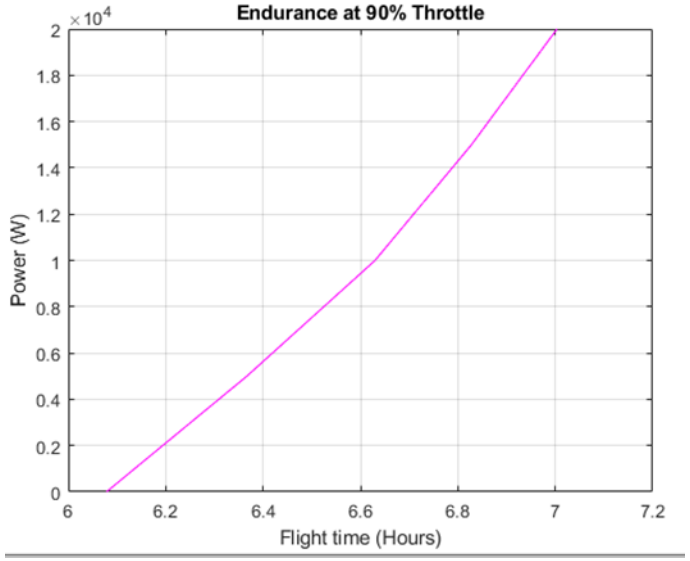
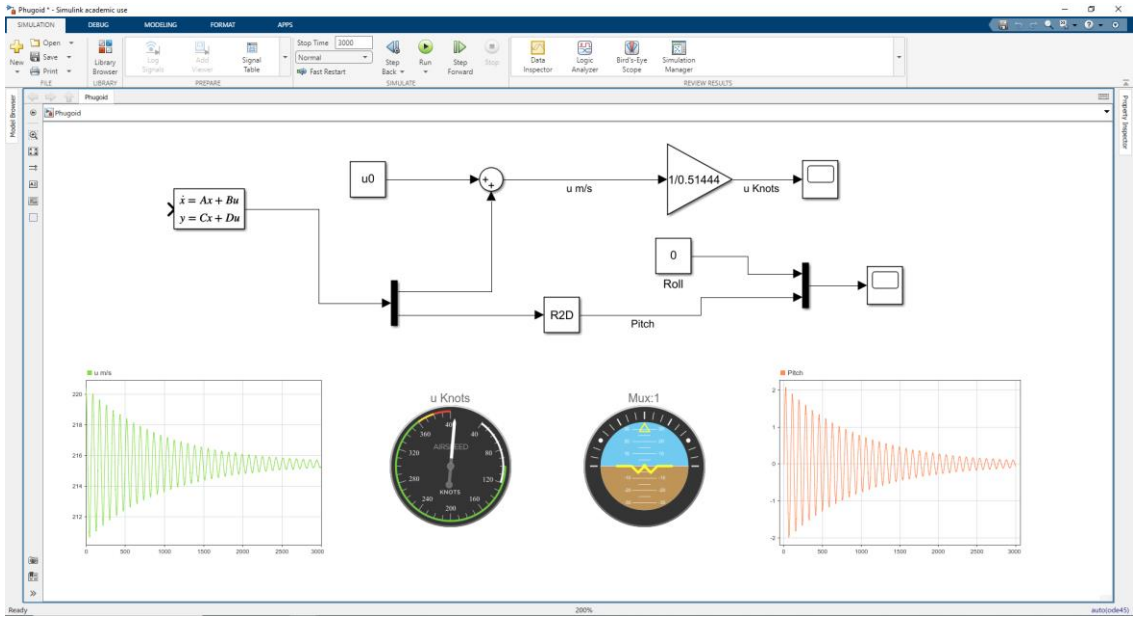


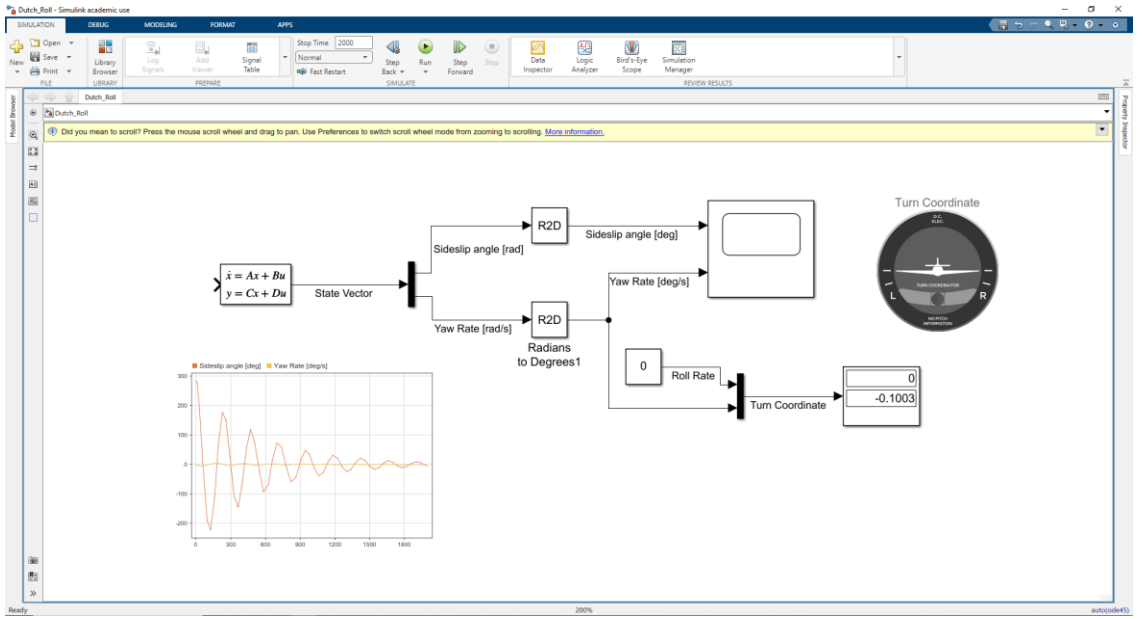
Figure 5

Phugoid



- A phugoid is a type of longitudinal motion in which the aircraft oscillates in pitch with a long period, typically lasting 20-60 seconds. It is a natural mode of motion for most aircraft and is caused by variations in lift and pitch moment as the aircraft climbs and descends.
- Where as my phugoid lasts 120 seconds but takes over 3000 seconds to stabilise.

Dutch Roll



- An aircraft Dutch roll is a type of lateral motion characterized by a combination of rolling and yawing oscillations, typically with a period of 4-8 seconds. It is caused by a combination of aerodynamic forces acting on the wings and vertical tail fin, and can be induced by disturbances such as turbulence or pilot inputs.
- This aircraft takes over1500 to stabilise.

Thank You

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